

DISCRETIZATION OF PARTIAL DIFFERENTIAL EQUATIONS

Supplemental – Non-examinable

Goal: to show how partial differential lead to sparse linear systems

- ▶ See Chap. 2 of text
- ▶ Finite difference methods
- ▶ Finite elements
- ▶ Assembled and unassembled finite element matrices

What does: 'Supplemental – Non-examinable' mean ?

- ▶ It simply means that this is less important material, that is ..
- ▶ .. not too critical to understanding the rest of the course
- ▶ No questions on it in quizzes
- ▶ There will be a few of these sets during this course. This is the first one.

Why study discretization of PDEs?

- ▶ PDEs are among the most important sources of sparse linear systems
 - ▶ Goal: to understand the structures of the problem and their connections with “meshes” in 2-D or 3-D space
 - ▶ Also: iterative methods are often formulated for the PDE directly – instead of a discretized (sparse) system.
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NOTE: It is useful to have an idea of how Finite Difference matrices are generated. For Finite Elements: goal is to unravel the related sparse computations to which they lead.

A typical numerical simulation

Physical Problem →

Nonlinear PDEs →

Discretization →

Linearization (Newton) →

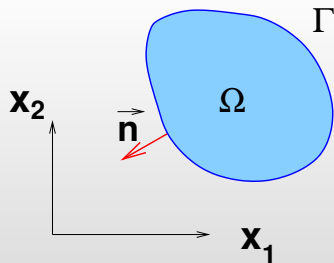
Sequence of Sparse Linear Systems $Ax = b$

Example: discretized Poisson equation

✦ Partial Differential Equation (PDE): Poisson's equation in 2 dimensions :

$$\frac{\partial^2 u}{\partial x_1^2} + \frac{\partial^2 u}{\partial x_2^2} = f, \text{ for } x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \text{ in } \Omega$$

where Ω = bounded, open domain in $\mathbb{R}^2$



✦ + boundary conditions:

Dirichlet: $u(x) = \phi(x)$

Neumann: $\frac{\partial u}{\partial \mathbf{n}}(x) = \mathbf{0}$

Cauchy: $\frac{\partial u}{\partial \mathbf{n}} + \alpha(x)u = \gamma$

- ▶ $\Delta = \frac{\partial^2}{\partial x_1^2} + \frac{\partial^2}{\partial x_2^2}$ is the Laplace operator or Laplacean
- ▶ How to approximate the Poisson problem shown above?
- ▶ Answer: discretize, i.e., replace continuum with discrete set.
- ▶ Then approximate Laplacean using this discretization
- ▶ Many types of discretizations.. will briefly cover Finite Differences (FD) and Finite Elements (FEM)

Finite Differences: Basic approximation formulas derived from Taylor series expansion:

$$u(x + h) = u(x) + h \frac{du}{dx} + \frac{h^2}{2} \frac{d^2u}{dx^2} + \frac{h^3}{6} \frac{d^3u}{dx^3} + \frac{h^4}{24} \frac{d^4u}{dx^4}(\xi)$$

Discretization of PDEs - Basic approximations

- ▶ Simplest scheme: forward difference

$$\begin{aligned}\frac{du}{dx} &= \frac{u(x+h) - u(x)}{h} - \frac{h}{2} \frac{d^2u(x)}{dx^2} + O(h^2) \\ &\approx \frac{u(x+h) - u(x)}{h}\end{aligned}$$

- ▶ Centered differences for second derivative:

$$\frac{d^2u(x)}{dx^2} = \frac{u(x+h) - 2u(x) + u(x-h))}{h^2} - \frac{h^2}{12} \frac{d^4u(\xi)}{dx^4},$$

where $\xi_- \leq \xi \leq \xi_+$.

Notation:

$$\delta^+ u(x) = u(x+h) - u(x)$$

$$\delta^- u(x) = u(x) - u(x-h)$$

✦ Operations of the type:

$$\frac{d}{dx} \left[a(x) \frac{d}{dx} \right]$$

are common [in-homogeneous media].

✦ The following is a 2nd-order approximation:

$$\begin{aligned} \frac{d}{dx} \left[a(x) \frac{du}{dx} \right] &= \frac{1}{h^2} \delta^+ \left(a_{i-\frac{1}{2}} \delta^- u \right) + O(h^2) \\ &\approx \frac{a_{i+\frac{1}{2}}(u_{i+1} - u_i) - a_{i-\frac{1}{2}}(u_i - u_{i-1})}{h^2} \end{aligned}$$

 **1** Show that $\delta^+ \left(a_{i-\frac{1}{2}} \delta^- u \right) = \delta^- \left(a_{i+\frac{1}{2}} \delta^+ u \right)$

Finite Differences for 2-D Problems

Consider the simple problem,

$$-\left(\frac{\partial^2 u}{\partial x_1^2} + \frac{\partial^2 u}{\partial x_2^2}\right) = f \quad \text{in } \Omega$$
$$u = 0 \quad \text{on } \Gamma$$

✦ $\Omega = \text{rectangle } (0, l_1) \times (0, l_2)$ and Γ its boundary.

Discretize uniformly :

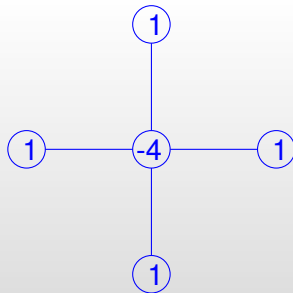
$$\begin{aligned} \text{uni-} \quad x_{1,i} &= i \times h_1 \quad i = 0, \dots, n_1 + 1 & h_1 &= \frac{l_1}{n_1 + 1} \\ x_{2,j} &= j \times h_2 \quad j = 0, \dots, n_2 + 1 & h_2 &= \frac{l_2}{n_2 + 1} \end{aligned}$$

Finite Difference Scheme for the Laplacean

- Use centered differences for $\frac{\partial^2}{\partial x_1^2}$ and $\frac{\partial^2}{\partial x_2^2}$ - with mesh sizes $h_1 = h_2 = h$:

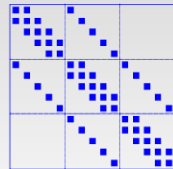
$$\Delta u(x) \approx \frac{1}{h^2} [u(x_1 + h, x_2) + u(x_1 - h, x_2) + u(x_1, x_2 + h) + u(x_1, x_2 - h) - 4u(x_1, x_2)]$$

The 5-point 'stencil:'



The resulting matrix has the following block structure:

$$A = \frac{1}{h^2} \begin{bmatrix} B & -I & \\ -I & B & -I \\ & -I & B \end{bmatrix}$$



Case: 7×5 grid

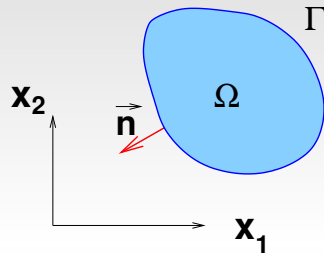
With

$$B = \begin{bmatrix} 4 & -1 & & & \\ -1 & 4 & -1 & & \\ & -1 & 4 & -1 & \\ & & -1 & 4 & -1 \\ & & & -1 & 4 \end{bmatrix}$$

Finite Element Method (FEM): a quick overview

Background: Green's formula

$$\int_{\Omega} \nabla v \cdot \nabla u \, dx = - \int_{\Omega} v \Delta u \, dx + \int_{\Gamma} v \frac{\partial u}{\partial \vec{n}} \, ds.$$



- ✦ The dot indicates a dot product of two vectors.
- ✦ ∇ = gradient operator. In 2-D: $\nabla u = \begin{pmatrix} \frac{\partial u}{\partial x_1} \\ \frac{\partial u}{\partial x_2} \end{pmatrix}$,
- ✦ Δu = Laplacean of u
- ✦ $\vec{n}$ is the unit vector that is normal to Γ and directed outwards.

- ◆ Frechet derivative:

$$\frac{\partial u}{\partial \vec{v}}(x) = \lim_{h \rightarrow 0} \frac{u(x + h\vec{v}) - u(x)}{h}$$

- ◆ Green's formula generalizes the usual formula for integration by parts

$$a(u, v) \equiv \int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} \left(\frac{\partial u}{\partial x_1} \frac{\partial v}{\partial x_1} + \frac{\partial u}{\partial x_2} \frac{\partial v}{\partial x_2} \right) dx$$

- ◆ Define

$$(f, v) \equiv \int_{\Omega} f v \, dx.$$

- ◆ With Dirichlet BC, the integral on the boundary in Green's formula vanishes $\rightarrow$

$$a(u, v) = -(\Delta u, v).$$

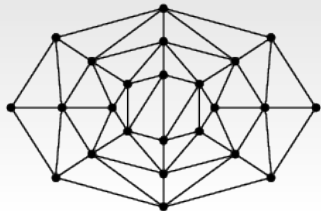
- ▶ Suppose we want to solve $-\Delta u = f$ in Ω + *Dirichlet BC*
- ▶ **Weak formulation** of the original problem: select a subspace of reference V of L^2 and then solve

$$\text{Find } u \in V \text{ such that } \underbrace{a(u, v)}_{=-(\Delta u, v)} = (f, v), \quad \forall v \in V$$

- ▶ Finite Element method solves this weak problem...
- ▶ ... by discretization

- ▶ The original domain is approximated by the union Ω_h of m triangles K_i ,

Triangulation of Ω :



$$\Omega_h = \bigcup_{i=1}^m K_i.$$

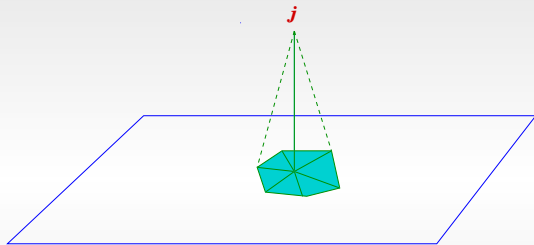
- ♦ Some restrictions on angles, edges, etc..

$$V_h = \{ \phi \mid \phi|_{\Omega_h} \in \mathcal{C}^0, \quad \phi|_{\Gamma_h} = \mathbf{0}, \quad \phi|_{K_j} \text{ linear } \forall j \}$$

- ▶ $\mathcal{C}^0$ = set of *continuous* functions
- ▶ $\phi|_X$ == restriction of ϕ to the subset X
- ▶ Let $x_j, j = 1, \dots, n$, be the nodes of the triangulation

- ✦ Can define a (unique) 'hat' function ϕ_j in V_h associated with each x_j s.t.:

$$\phi_j(x_i) = \delta_{ij} = \begin{cases} 1 & \text{if } x_i = x_j \\ 0 & \text{if } x_i \neq x_j \end{cases}.$$



- ✦ Each function u of V_h can be expressed as $u(x) = \sum_{j=1}^n \xi_j \phi_j(x).$ (*)

- ▶ FEM approximation $\equiv$ Galerkin condition for functions in V_h :

$$\text{Find } \mathbf{u} \in V_h \text{ such that } \mathbf{a}(\mathbf{u}, \mathbf{v}) = (\mathbf{f}, \mathbf{v}), \quad \forall \mathbf{v} \in V_h$$

- ▶ Express $\mathbf{u}$ in the basis $\{\phi_j\}$ (see *), then substitute above. Result:

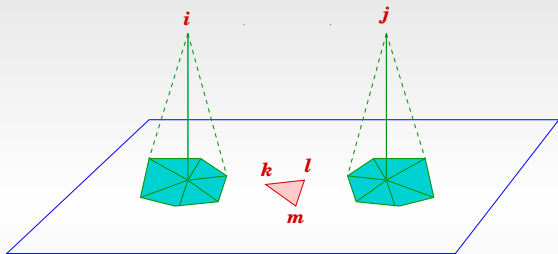
▶ Linear system $\sum_{j=1}^n \alpha_{ij} \xi_j = \beta_i$ where: $\alpha_{ij} = \mathbf{a}(\phi_j, \phi_i), \quad \beta_i = (\mathbf{f}, \phi_i).$ =

- ▶ Above equations form a linear system of equations $\mathbf{Ax} = \mathbf{b}$

- ▶ $\mathbf{A}$ is *Symmetric Positive Definite*

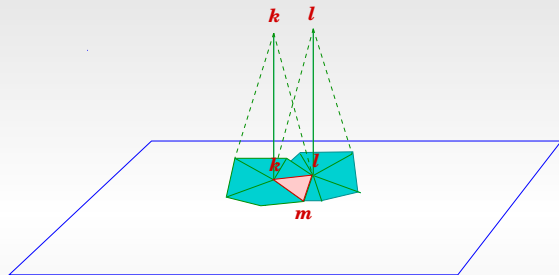
 **2** Prove that $\mathbf{A}$ is *Symmetric Positive Definite*

The Assembly Process: Illustration



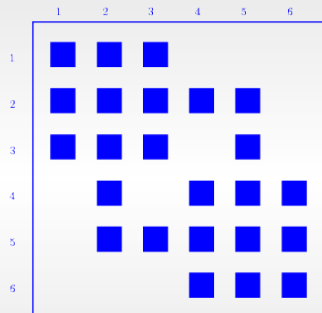
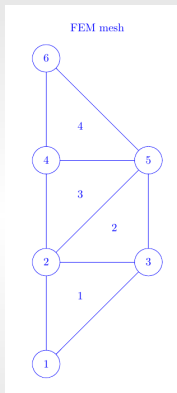
If triangle $K \notin$ support domains of both ϕ_i and ϕ_j then $a_K(\phi_i, \phi_j) = 0$

► So: $a_K(\phi_i, \phi_j) \neq 0$ iff $i \in \{k, l, m\}$ and $j \in \{k, l, m\}$.

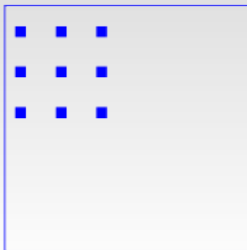
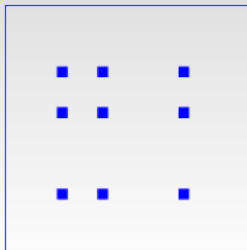
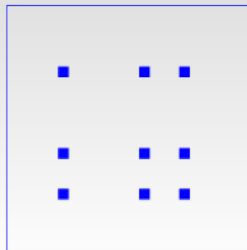


If triangle $K \in$ *both* nonzero domains of ϕ_i and ϕ_j then $a_K(\phi_i, \phi_j) \neq 0$

The Assembly Process



Small finite element mesh and pattern of the corresponding assembled matrix

$A^{[1]}$  $A^{[2]}$  $A^{[3]}$  $A^{[4]}$ 

Element matrices $A^{[e]}$, $e = 1, \dots, 4$ for FEM mesh shown above

- Each element contributes a 3×3 submatrix $A^{[e]}$ (spread out)
- Can also use the matrix in un-assembled form - To multiply a vector by A for example we can do:

$$y = Ax = \sum_{e=1}^{nel} A^{[e]} x = \sum_{e=1}^{nel} P_e A_{K_e} (P_e^T x)$$

- ▶ Can be computed using the element matrices A_{K_e} - no need to assemble
- ▶ The product $P_e^T x$ gathers x data associated with the e -element into a 3-vector consistent with the ordering of the matrix A_{K_e} .
- ▶ Advantage: some simplification in process
- ▶ Disadvantage: cost (memory + computations).

Resources: A few matlab scripts

✦ These (and others) will be posted in the matlab folder of class web-site

```
>> help fd3d
```

```
function A = fd3d(nx,ny,nz,alpx,alpy,alpz,dshift)
```

NOTE nx and ny must be > 1 -- nz can be $= 1$.

5- or 7-point block-Diffusion/conv. matrix.

A stripped-down version is `lap2D(nx,ny)`

```
>> help mark
```

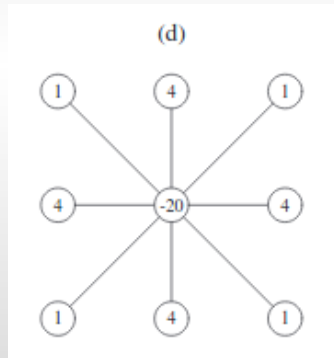
```
[A] = mark(m)
```

generates a Markov chain matrix for a random walk
on a triangular grid. A is sparse of size $n=m*(m+1)/2$

3 Explore A few useful matlab functions

- * `kron`
- * `gplot` for plotting graphs
- * `reshape` for going from say 1-D to 2-D or 3-D arrays

4 Write a script to generate a 9-point discretization of the Laplacean.



- ▶ `generateMesh(model, ...)` Generates the mesh
- ▶ `results = pdesolve(model, ...)` solves the PDE
- ▶ `pdeplot(model, ...)` plots solution
- ▶ Also `assembleFEMatrices(model, ...)` assembles the FEM problem, [returns K and M in a structure]

 **5** Follow the example in the documentation and get an understanding of the functions that are called.